

1. Welcome to SBS Step Browser

SBS turns a CAD assembly into a narrated, step-by-step illustration: every step is a complete picture of the scene — which parts are visible, where they are, what colour they wear, where the camera stands, what is drawn on top — and SBS animates the journey from one step to the next, speaks the narration, and renders the whole thing to video.

What SBS is for

Assembly and service instructions, training videos, exploded-view walkthroughs, sales demos of mechanical products. You load a model once, then *author steps*: hide a cover, move a screw out, point a camera, write a sentence. SBS handles the in-between motion, the voice, the subtitles, the header, the translation into other languages and the final MP4.

The five ideas you need

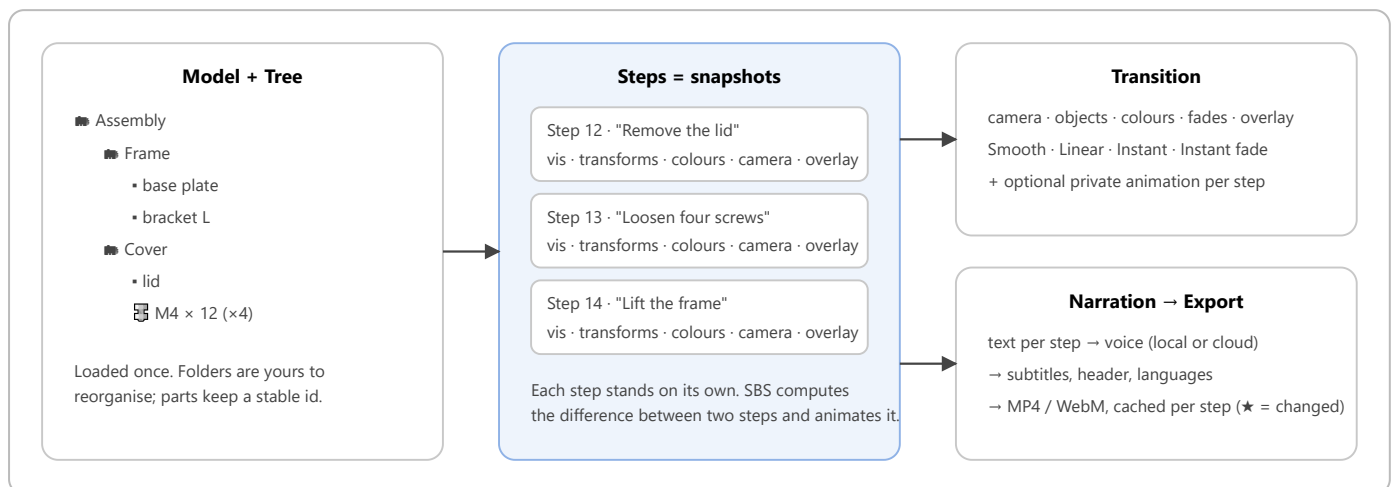


Figure 1 — Model and tree on the left, steps (self-contained snapshots) in the middle, transitions and export on the right.

Project

One `.sbsproj` file inside its own folder, plus sidecar folders for audio, render cache, media, languages, exports and backups. The project folder is meant to be moved or copied as a unit.

Model and tree

The CAD geometry (STEP, IGES, OBJ, GLB/gLTF, STL) shown as a tree of folders and parts in the **Tree** tab. You can regroup, rename, archive, add hardware, cables, primitives, notes.

Step

A complete snapshot of the scene: visibility, transforms, colours, camera, overlay, narration. Steps are grouped into *chapters*. A step never depends on the steps around it — you can jump anywhere and the scene is correct.

Transition

What the viewer sees between two steps. SBS diffs the two snapshots and animates the difference according to the step's easing (Smooth / Linear / Instant / Instant fade) and an optional private animation.

Export and cache

Each step is rendered once into a per-step cache. When you edit something, only the steps it touches get a ★ star and are re-rendered. Everything else is stitched from cache.

How to use this manual

The left column is a searchable table of contents. Type anything — a button label, a shortcut, a symptom — and matching topics appear with a snippet. Press to jump to the search box, to clear it. The  **Theme** button switches light and dark.  **Print** produces a PDF with one chapter per page section.

Labels written like **Tree** are the exact words you see in the app. Keys are written like + . Menu paths are written **Edit ▶ Languages....**

2. Getting started

Install and launch

Run `SBS Step Browser Setup <version>.exe`. The installer lets you choose the folder, creates a desktop and Start-menu shortcut, and does not touch your projects on uninstall. SBS is fully offline: the text-to-speech voices, the CAD converters and the video encoder ship inside the installer. Cloud text-to-speech and machine translation are optional and need an internet connection plus an API key.

First launch is slower: the local voice engine warms up in the background (a few hundred megabytes of models are loaded once per session). You can start working immediately; narration playback becomes available when the status bar stops reporting the warm-up.

Your first project in ten minutes

1. **File** ▶ **New** (`Ctrl + N`), then **File** ▶ **Save As...** to give the project a folder. Save early — the render cache, audio and backups live next to the project file.
2. Open the **Files** tab and click **Import**. Pick a STEP, IGES, OBJ, GLB or STL file. STEP and IGES are tessellated on import; large assemblies take a while the first time and are then cached as fast-loading `.sbsobj` files.
3. Look at the **Tree** tab. Rename or regroup folders so that they match how you want to explain the assembly (see chapter 4).
4. Orbit with the left mouse button, pan with the right button or middle button, zoom with the wheel. Press `F` to fit the selection (or everything) in view.
5. In the steps panel at the bottom, add a step. Hide a part (eye icon in the tree, or the Visibility commands in the right-click menu), move another with the gizmo, press `C` to capture the camera into the step. Add the next step and repeat.
6. Type the narration for each step in the text field under the viewport. Press the  button to hear it.
7. Press the  button to play the sequence. Adjust transition easing per step if something should snap instead of glide.
8. Open the **Export** tab, choose size, frame rate and format, and click **Start Export**. Steps are rendered once into the cache; later exports only re-render starred steps.

What a project folder looks like

MyProduct/	
MyProduct.sbsproj	the project — single source of truth
MyProduct.he.sbslang.json	one language pack per extra language (see chapter 13)
audio/<lang>/	synthesised narration clips per language and voice
render/<lang>/	per-step render cache, frame records, alpha masks
media/	images and video clips used by overlays
exports/	finished MP4 / WebM / .sbsproc files
backups/	automatic backups (Settings ► Auto-backup)
model.sbsobj	fast-load copies of tessellated CAD (offered after the first STEP load)

Figure 2 — A self-contained project folder. Move or copy the whole folder; never move only the `.sbsproj`.

Older projects. Projects saved before the folder layout existed keep their files next to the project (for example a `_rendercache` folder). SBS reads both layouts; new material is written to the new one.

Saving, autosave, backups

`Ctrl` + `S` saves, `Ctrl` + `Shift` + `S` saves as. Saves are atomic — the previous file is only replaced once the new one is completely written. **Settings** ► **Auto-backup** keeps timestamped copies in `backups/`; **Back up now** makes one immediately. Closing the window with unsaved changes asks you first.

Rule of thumb. The project file is the truth. Anything you see in the viewport is rebuilt from it when a project loads, so a reload (`Ctrl` + `R` in the app window) is always a safe way to make the live scene match the data.

3. The interface

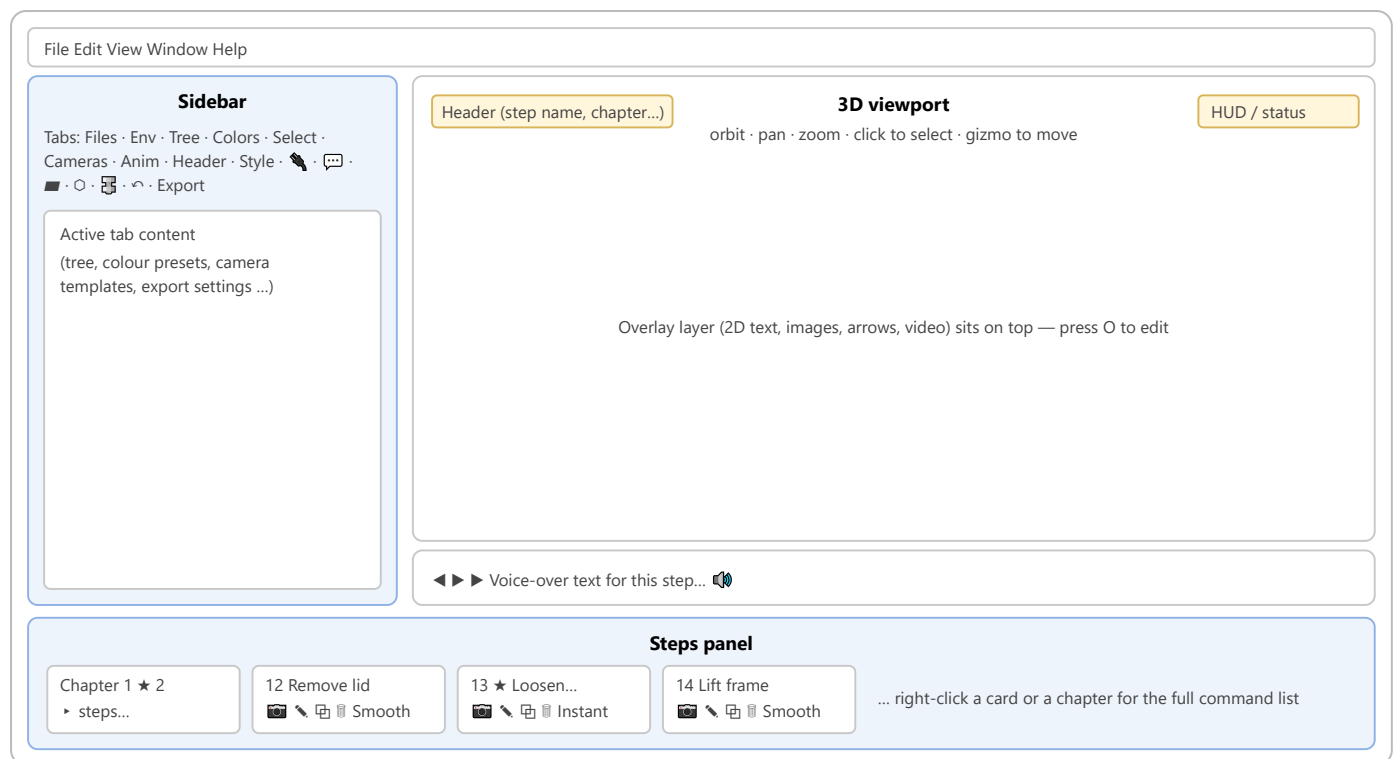


Figure 3 — Layout map: menus on top, sidebar on the left, the 3D viewport with the header and overlay layer, the narration bar, and the steps panel along the bottom.

Menus

Menu	Items
File	New Ctrl + N · Open... Ctrl + O · Save Ctrl + S · Save As... Ctrl + Shift + S · Browse Assets Ctrl + B · Settings Ctrl + , · Exit
Edit	Undo / Redo · Cut / Copy / Paste / Select All · Recover stuck inputs · Apply Camera Template... · Languages... · Model source transform...
Tools	Brand... · Review form... · Document... · Title Manager... · Unify Constant Titles · Clean Up Unused Constant Titles · Rebuild Cascade · Clean up folders... · Scan steps for what is in frame...
View	Zoom In / Zoom Out / Reset Zoom (interface scale) · Toggle Full Screen · Developer Tools Ctrl + Shift + I · Reload Ctrl + R · Hard Reload Ctrl + Shift + R
Window	Minimize · Close
Help	SBS Step Browser Help

Each Edit command is explained in the chapter it belongs to: camera templates in chapter 8, titles and languages in 12 and 13, Rebuild Cascade / Clean up folders / Model source transform in chapter 20, the frame scan in chapter 19.

Sidebar tabs

Tab	What lives there	Chapter
Files	Import models, the list of loaded assets, browse / relink, missing-asset warnings.	4
Env	Background, grid, outline, theme, lights, production-render look (AO, reflections, grade).	18
Tree	The assembly hierarchy: folders, parts, hardware, cables, primitives, shapes, notes. Filters, locks, visibility.	4
Colors	Colour presets and material properties; assign to the selection; highlight colour.	6
Select	Saved selection groups you can load, update or delete.	5
Cameras	Named camera templates; bind steps to them; go to / update / refit.	8
Anim	Animation presets, AL1 / AL2 timing variables, the private-animation editor.	9
Header	The on-screen header: step number / name, chapter, text, image, progress bar, subtitle.	11
Style	Text styles and shape styles for the overlay.	10
	Cables and sockets.	15
	Notes attached to parts, and note templates.	17
	Flat shapes traced from model faces, shape groups, textured planes.	16
	Primitives (boxes, cylinders and so on) placed on faces.	16
	Hardware templates (screws, nuts, washers) and placement.	14
	Undo history — jump back to any earlier state, or clear it.	5
Export	Resolution, frame rate, hold, voice, format, range; the render cache; start the export.	19

Viewport basics

- **Orbit** left-drag, **pan** right- or middle-drag, **zoom** wheel. **F** fits the selection; with nothing selected it fits everything. **Cameras** ▶  **Refit camera** recomputes orbit distance and pivot after a model rescale.
- **Work camera** (**W**): a scratch camera you can move freely without touching the step's saved camera. Press **W** again to return to the step camera. **C** captures the current view into the active step.
- **Select** by clicking a part. Click again on the same spot to climb one level up the tree (part → folder → model). Double-click a locked folder to temporarily unlock it; it relocks when nothing inside is selected any more. **Shift** adds to the selection, **Ctrl** toggles, **Alt** inverts within the clicked unit. The cursor shows a small badge for the active modifier.

- **Gizmo** appears on the selection: drag the arrows to move, the rings to rotate. **L** switches the gizmo between local and world axes. Type exact values in the numeric fields next to it.
- **Overlay edit** (**O**) switches the mouse to the 2D layer on top of the viewport (see chapter 10). **X** toggles overlay x-ray so you can see the model through the overlay while editing.
- **Global mode** (**Space**): while it is on, a transform you make carries into every step, not only the active one (see 5.5).
- **Esc** cancels a placement tool or picker, closes a popup, or deselects.

Narration bar and playback

Below the viewport: ◀▶ step back and forward, ▶ plays the sequence from the active step, the text field holds the voice-over for this step, 🔊 speaks it. ◀ / ▶ also move between steps when the focus is not in a text field.

Steps panel

Chapters are the grey headers; steps are the cards. A card shows its number, name, easing, and small buttons: 📷 update this step's camera from the current view, ✎ edit this step's private animation, 📄 duplicate, 🗑 delete. A ★ means the step changed since it was last rendered; a chapter header shows how many of its steps are starred. Right-click anything for the full command list (chapter 7).

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the steps panel with a chapter header, starred cards and the easing dropdown. File:

`docs/manual/img/steps-panel.png`

4. Models and the tree

Importing models

Files ▶ Import accepts `.step / .stp`, `.iges / .igs`, `.obj`, `.glb / .gltf`, `.stl` and SBS's own `.sbsobj`. Several models can live in one project; each becomes a root node in the tree.

- **STEP / IGES** are tessellated by the bundled converter. **Settings ▶ CAD import** sets the quality. After the first successful load SBS offers to save a `.sbsobj` next to the project — the exact same tessellation, loading in seconds instead of minutes. Accept it.
- **GLB / glTF / FBX / OBJ** arrive with their own hierarchy and, sometimes, baked transforms. SBS flattens what it must and records each mesh's resting pose so that steps land on it exactly.
- **Units. Edit ▶ Model source transform...** lets you scale, rotate or offset a whole model at the source level (Metric / Imperial helpers, per-axis reset). Do this once, before authoring steps.

Keep the source files. The project stores the tree and identities, not the geometry. On load, SBS looks for the asset at the recorded path (absolute or relative to the project). If it is missing you get the asset check dialog: **Browse...** to relink, **Override** to load anyway with a placeholder, **Clear All Warnings** once you have fixed the paths. **File ▶ Browse Assets** (`Ctrl + B`) shows the same list any time.

Reading the tree

The **Tree** tab lists the hierarchy. Icons tell the node type: folder, mesh,  hardware, cable, primitive, flat shape, note. Left of every row: the eye (visible in this step), the lock (locked folders select as one unit and refuse edits inside). The filter bar narrows the list: **Sel** only selected objects, **Arch** only archived, **HW** only hardware, **Vis** only visible, **Hid** only hidden, **Fold** only containers (selected parts stay visible). Type in the search field to filter by name. **Unhide all** shows everything in the current step.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the Tree tab with filters, eyes and locks. File: `docs/manual/img/tree-tab.png`

Organising folders

Drag rows to reparent. Right-click a node for:

Command	What it does
 Rename	Renames the node. Names are what the header, notes and diagnostics show.
 + New Folder Inside	Creates an empty folder under the node.
 Move Selected Here	Reparents the current selection into this folder.
 Collapse	Collapses the branch in the list.
 Clean redundant folders	Removes single-child wrapper folders below this node (CAD exports love those). The project-wide version is Edit ▶ Clean up folders... , which previews the changes, then offers Save first, then remove .
 Delete Empty Folder / Delete Folder (empty first)	Folders with content are protected — empty them before deleting.
 Delete assembly...	Removes a whole loaded model and every reference to it. Asks first.
 Copy tree	Copies the branch as text (for reports or for pasting into a message).
 Archive /  Unarchive	Archived parts drop out of every step and out of the picture, without being deleted. Use it for the fasteners you never want to explain. The Arch filter shows them.
 Isolate /  Un-isolate	Temporarily shows only the node (a viewing aid — it does not change the step).
 Make transformable	Turns a raw imported mesh into a node with its own pivot and transform so it can be moved in steps.

Visibility across steps

The eye icon changes visibility for the *active* step only. To cascade, right-click ▶  **Visibility**:

- ▶  Show on all following steps / ▶  Hide on all following steps
- ◀  Show on all previous steps / ◀  Hide on all previous steps
- ⦿  Show ONLY on this step

Hiding a folder hides everything inside. A part that is hidden by an ancestor shows a dimmed eye. During a transition a part that becomes hidden fades out; one that appears fades in (the fade length is the visibility channel of the animation, chapter 9).

Replace-models and following

Two ways to say "this object should look different in some steps":

-  **Convert to Replace-Model** makes the node a holder for alternative geometry. + **Add to replace...** adds a variant; each step can show one of them.   **Remove from replace-model** undoes it.

-  **Follow object...** makes object A ride along with object B: wherever B's folder moves in later steps, A comes along, while A keeps its own visibility and colour.  **Stop following...** releases it. Following is the recommended way to attach hardware, labels or tools to a moving part.

Notes on a part

 **Add Note...** attaches a callout to the node;  **Notes on this folder (n)** lists them, with  **Reposition Note...** and  **Delete Note**. See chapter 17.

5. Selecting, moving, pivots

Selection rules

- Single click selects the smallest thing under the cursor (a part). Clicking the same spot again selects its folder, then the model — a climb up the tree. In the tree, click a row or Ctrl / Shift-click for multi-select.
- A **locked folder** (🔒 in the tree) behaves as one object: clicking any member selects the whole folder. Double-click it to unlock temporarily and edit a part inside; it relocks by itself once nothing inside is selected.
- **Colors** ▶ **Select All Meshes** / **Deselect All** do what they say. The Colors tab can also select every part wearing a colour.
- **Select** tab:  **Save...** stores the current selection as a named group;  **Load...** re-selects it;  **Update** overwrites it from the current selection;  **Delete** removes it. Groups survive save/load and are handy for "all the screws of the cover".

Moving with the gizmo

The gizmo appears at the selection's pivot. Drag an axis arrow to move along it, a plane handle to move in a plane, a ring to rotate. L toggles **local** (the part's own axes) and **world**. The numeric fields accept exact values — type a number and press Enter. Every gizmo edit is undoable (Ctrl + Z).

What you move is stored in the *active step* only. Go to the next step and the part is back where that step says. That is by design: a step is a full snapshot. To move a part in many steps at once use Global mode or Group for global edit.

Transform commands (right-click)

Command	Use
 Copy Transforms /  Paste Transforms	Copy a part's position and rotation in this step and paste onto another part (or the same part in another step).
 Reset Move /  Reset Rotation /  Reset All Transforms	Send the part back to its imported pose in this step.
 Global transform...	Apply one numeric move / rotation to the selection across all steps at once.
 Align to surface...	Click a face on another part; the selection's base face is laid tangent onto it.
 Align by 3 points (concentric)...	Pick three points on a circular edge of the target and three on the moving part: the axes are matched (screws into holes, shafts into bearings).
 Fit To	Frames the node in the viewport.

Pivots

Rotation happens around the pivot.  **Copy Pivot** /  **Paste Pivot** move a pivot between parts, and  **Paste Pivot** → **selected steps (n)** pushes it into a step selection.  **Pivot Center via 3 Points...** sets the pivot on the centre of a circle you click three times (hinges, shafts).  **Snap Pivot to Surface...** puts it on a clicked face. When a part rotates around a pivot between two steps, SBS animates the arc, not a straight line — also when the part sits inside a moving parent folder.

Global mode (Space)

Press Space to toggle Global mode (the HUD shows it). While it is on, a move or rotation you make is applied as a *delta* to the selected objects in *every step* the moment you deselect. Use it to fix an assembly-wide placement error without touching each step. Press Space again to go back to per-step editing.

Group for global edit

 **Group for global edit...** wraps scattered parts into a helper folder so that one transform can drive them together across all steps; it shows a dry-run of what it will change before doing it. Pair it with Global mode.

Undo and the history tab

Every edit — tree, transforms, colours, steps, overlay, cables, hardware, settings — goes through one undo stack. `Ctrl + Z` / `Ctrl + Y`, or open the ↶ tab to see the list and jump back several entries at once. 🗑

Clear history frees memory on very long sessions. Loading a project clears the history.

6. Colours and materials

Colour presets

The **Colors** tab holds a list of named presets. Every part wears exactly one preset per step; the preset is what steps store, not raw RGB. Change a preset and every part wearing it changes everywhere. + **Add Color** creates one; double-click to rename; the ★ **Set as Default** preset is the one new parts get.

- **Assign to Selected** applies the highlighted preset to the selection in the active step (a visible colour change between two steps animates as a cross-fade).
- ↺ **Revert to Default** restores each selected mesh to its default colour.
- **Highlight Color** is the colour used for selection highlighting and for parts that a note points at.
- The sort buttons float the colours used by currently visible objects (or by the selection) to the top; the rest turn grey but stay clickable.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the Colors tab with presets and material sliders. File: `docs/manual/img/colors-tab.png`

Material properties per preset

Property	Effect
Solid color	The base colour.
Opacity	Classic transparency (0–1). Transparent parts are drawn after solid ones.
Solidness	X-ray strength. Below 1 the part becomes a ghost through which other parts show — this is the usual way to reveal an interior without hiding the cover.
Metalness / Roughness	Physically based shading: metals reflect the environment, rough surfaces spread the highlight.
Reflection Intensity · Reflective (SSR contact reflections)	Screen-space reflections for floors and polished plates. Cost is real; use on a few presets.
Strip textures (pure solid color)	Ignore the texture that came with the mesh and use the flat colour.
Outline: Global default / Always on / Always off	Whether parts with this preset get the silhouette outline from the Env tab.
Flat mirror	Turns a planar face into a true mirror (console-driven feature; see chapter 21).

Slider drags are batched into one undo entry per drag.

Colours and steps

A colour assignment is part of the step snapshot. To make a part red only in step 12, assign it in step 12. To make it red from step 12 on, assign it in step 12 and use the step's copy/duplicate to carry it forward, or assign it in each later step. Changing a *preset*'s values (its RGB or solidness) affects every step and stars only the steps where a part wearing that preset is actually in frame (see 19.4).

Deleting a preset is refused while a part still wears it, and the default preset cannot be deleted. Assign the parts elsewhere first.

7. Steps and chapters

What a step stores

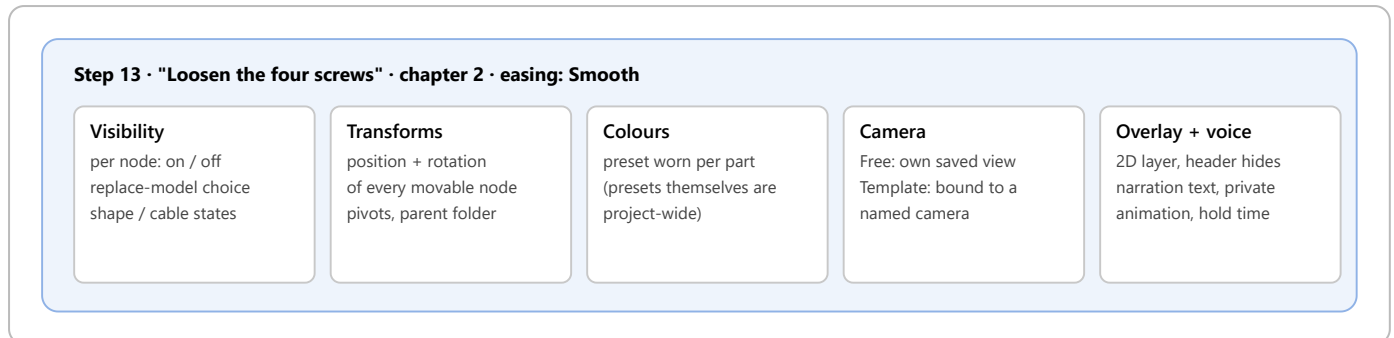


Figure 4 — Everything a step needs to stand alone. Jumping to any step rebuilds exactly this picture.

Creating and arranging steps

- **Add** a step with the panel's add button: the new step starts as a copy of the active one, so you only change what differs. on a card duplicates it in place.
- **Reorder** by dragging cards; drag into another chapter to move it. Multi-select with Ctrl / Shift - click, then drag, copy or delete the whole selection.
- **Chapters** group steps and give the header its chapter number and name. Right-click a chapter header for **Rename...**, **Copy**, **Delete**, selection helpers and **Paste steps into chapter**.
- **Step groups**: **Convert to step group** turns a step into a container whose children play as one continuous clip (one narration, several motions). **Ungroup step** reverses it.
- **Delete** with , Delete or Backspace when the card has focus. The last step cannot be deleted.

Step context menu

Command	What it does
 Rename...	Step name (shown in the header and in exports).
 Auto-name all (by chapter) ·  Auto-name all (from narration)	Name every step "Chapter n – step m" or from the first words of its narration.
 Duplicate ·  Copy ·  Paste under	Copy carries the complete snapshot, including overlay and narration. Paste inserts after the clicked step; with several copied, all of them.
 Import steps from another project...	Opens a second <code>.sbsproj</code> and lets you pick steps to bring in. Works best when both projects load the same CAD (part identities match); missing parts are reported.
 Copy overlay ·  Paste overlay — Replace /  Add on top	Move the 2D layer between steps without touching the 3D snapshot.
 Update step camera	Store the current viewport camera in this step (same as ). With several steps selected, all of them.
 Re-render cache segment	Force this step (or the selection) to be rendered again at the next export.
☆ Clear star (treat as rendered) · ☆ Clear all stars	Tell SBS the cached clip is fine even though something changed (19.4).
< cable > · < verb > · < socket > · State n	Cable actions recorded on this step (plug / unplug); ✕ Remove this action deletes one.
☒ Select all steps in chapter · Select all steps & chapters · ⇌ Invert selected	Selection helpers for bulk operations.
 Delete	Removes the step (or the selection). Undoable.

The transition row

Under the active card sits the transition row: **easing**, **camera mode** (Free / Template), **AL1 / AL2** durations and **Smooth reparent jumps**.

Easing	What the viewer sees when arriving at this step
Smooth	Eases out of the previous step and into this one. Default.
Linear	Constant speed. Good for conveyor-like motions and camera slides that continue into the next step.
Instant	The step arrives all at once: a purple INSTANT BLOCK is pinned to the top of the step's animation and holds every channel (camera, objects, colours, visibility, overlay, cables...). Use it for cuts.
Instant fade	Same, but the picture dissolves to nothing and back over AL1 while it cuts.

AL1 and **AL2** are the two animation-length variables in milliseconds. Presets reference them instead of fixed numbers: AL1 is the default home of the camera channel, AL2 of the object, visibility and colour channels. Change them on the step to speed up or slow down that one transition; change the defaults in the **Anim** tab.

Smooth reparent jumps: when an object changes folders between two steps but barely moves, hold it still instead of letting it swing through a large arc. The threshold is the distance its visual centre may move before the arc is animated normally.

Instant on the first step of a chapter. Works like any other step. If an old project shows a strange half-transition on such a step, re-pick the easing once from the dropdown — that rewrites the instant block in the current format.

Steps that are videos

A step can also be a rendered clip imported from a render cache (marked "This step IS a rendered video") or contain a video overlay playing alongside the scene. Video steps are exported as-is and cannot be edited in 3D; edit the original project instead.

8. Cameras

Free cameras

By default a step's camera is **Free**: whatever view you capture with **C** (or  on the card) is saved in that step. Moving the viewport does not change the saved camera until you capture again. Press **W** to enter the work camera — a scratch view for inspecting the model — and **W** again to snap back to the step's saved camera.

Camera templates

The **Cameras** tab holds named views. + **Save current view as template** creates one from the viewport. For each template: ► **Go To** moves the live camera there (bindings untouched),  **Update** sets the template to the current view *and* binds the active step to it,  **Lock** protects it from accidental updates, ★ **Set as Default** makes it the camera new steps get,  **Delete** removes it (bound steps become Free with the last view baked in).

Switch a step's camera mode to **Template** in the transition row and pick a name: from then on the step follows the template, and updating the template updates every step bound to it. **Edit** ► **Apply Camera Template...** binds a whole range of steps at once.  **Refit camera** recomputes distance and pivot to frame the entire scene — useful after a model rescale.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: camera templates with Go To / Update / Lock. File: `docs/manual/img/cameras-tab.png`

How the camera animates

Between two steps the camera interpolates position, target and zoom over AL1 (or over the camera channel of a private animation). With **Instant** easing it cuts. Orbit pivots and pull-outs stored on a step are honoured. The export renders exactly what the step camera sees at the export aspect ratio; the viewport may show a little more or less on the sides.

Settings ► **Camera** holds the defaults for new projects (field of view, orbit behaviour, default background).

9. Transitions and the animation editor

Channels

A transition is made of channels, each with its own duration. SBS animates whichever channels actually changed between the two steps:

Channel	Animates
camera	Camera position, target, zoom (home: AL1).
obj	Object moves and rotations, pivot arcs, reparent jumps (home: AL2).
visibility	Fade-in / fade-out of parts that appear or disappear (home: AL2).
color	Cross-fade between colour presets (home: AL2).
overlay / overlays	The 2D layer's fade-out of the old overlay and fade-in of the new one.
insert	Hardware insertion (screws driving in, with the reposition pre-step).
cable	Cable morphs and socket plug / unplug.
shape	Flat shapes and primitives appearing, moving, resizing.
notes	Note callouts appearing and following their parts.
narration	Where the voice-over starts relative to the motion.
pause	A time spacer with no channels — a deliberate hold inside a transition.

Animation strings and presets

The **Anim** tab shows the default preset as a string such as `camera(AL1), color(500), visibility(AL2), obj(AL2)`. Channels listed together in one block run at the same time; blocks separated by a phase run one after the other. Numbers are milliseconds; `AL1` / `AL2` resolve at transition time from the step (or the project defaults).

- **+ New Preset** creates a preset; the ★ marks the **Project default** used by steps that have no private animation.
- Presets can be imported from your personal collection (kept in user settings, across projects) and exported to it.
- The three **search zones** filter the preset list by dragging channel chips into them; **Clear all 3 search zones** resets.

Private animation per step

Press  on a card to open the editor for that step only. It shows the phases as blocks; drag channel chips between blocks to reorder what happens first, add an empty time block or a pause block, type a custom duration or pick AL1 / AL2, and remove blocks with  **Remove time block**. The raw string is shown underneath and can be edited by hand.

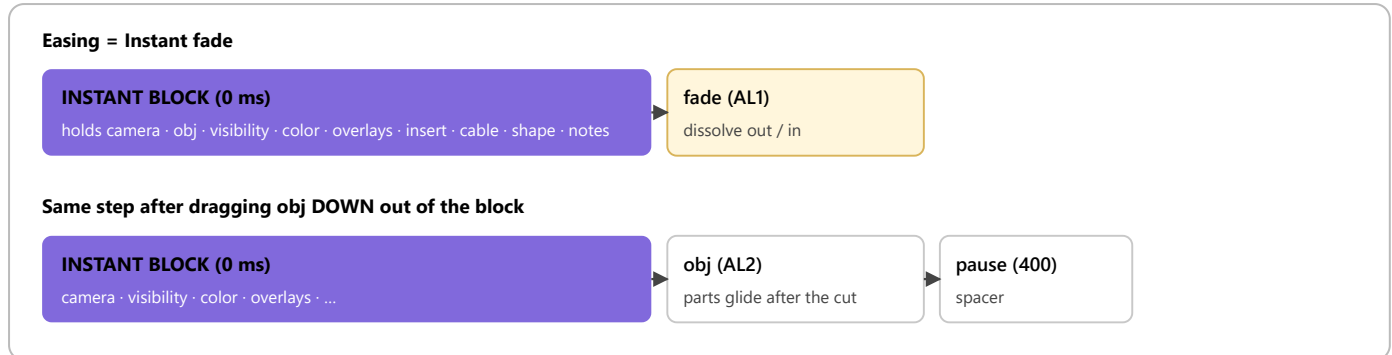


Figure 5 — Instant easing pins a purple block that holds every channel. Drag a chip down out of it and that channel animates after the cut.

Order of precedence. A step's private animation wins over the project default. The easing dropdown edits the private animation (it inserts or removes the instant block). Deleting the private animation returns the step to the default preset.

Holds and narration timing

After the transition the step is *held* while the narration plays. In the exported video the hold is the narration length plus **Export** ▶ **Step hold (ms)**. A pause *inside* a transition is authored with the pause channel, not with the hold.

10. The overlay layer

Entering overlay edit

The overlay is a 2D canvas locked to the picture: text boxes, images, arrows, rectangles, zoom insets, video clips, 3D-anchored arrows. Press **O** (or the overlay button) to edit it; the 3D mouse is suspended until you exit (**O** again, or **Exit overlay** in the prompt that appears when you leave the step). **X** toggles x-ray so the model stays visible under dense overlays while editing (its button sits in the helper row under the overlay bar, next to the magnet). Each step has its own overlay; use **Copy overlay** / **Paste overlay** on the card menu to move it.

Tools

Tool	Notes
Text	Double-click to edit in place: the editor shows the box exactly as it will render — template font and size, bold, effects, fill, rotation. The text toolbar sets font, size, colour, alignment, effects (shadow, outline) and a saved text style from the Style tab. The size list ends with Custom... : type any size (78, 150, 22.5 — pixels of the export frame), Enter applies it to the selected characters and the value joins the list. Whatever you last picked becomes the look of the next new text box. Hebrew and mixed Hebrew/English lines read correctly by default (each paragraph follows its first letter); if a line opens with a number or an English term and comes out wrong, right-click ▶ ¶ Text direction ▶ Right-to-left forces it for the whole box.
Image	PNG / JPG / SVG from disk; copied into media/ . Right-click ▶ 🖼️ Change image... , ↶ Reset to natural size , 🔍 Add zoom (an inset crop), 🔍 Zoom level 1× ... Custom.
Shapes and arrows	Rectangle, line / arrow (arrowhead on tail / head), corner radius, fill and stroke from the shape toolbar or a shape style . ➕ Reposition HEAD... / TAIL... re-anchor an arrow by clicking.
 3D arrow	An arrow whose head is pinned to a point on the model: it follows the part across steps and camera moves.
Video	A clip from disk (never copied into the project file). Right-click ▶ 🎞️ Trim & audio... for in/out points and mute. The clip plays in the viewport and is included in the export.
Image sequence	A flipbook of frames with timing (the sequence editor: + Add frame... , seconds per frame, Clear sequence).

Common commands: **📄 Copy**, **📄 Paste**, **📄 Paste in place**, **📄 Duplicate**, **🗑️ Delete**, **📏 Arrange** (Bring to front / forward, Send backward / to back). Arrow keys nudge; **Shift** while rotating levels to 45° / 90°; the rotation field accepts a typed angle.

Selecting with a box. Press on an *empty* part of the picture and drag: a box follows the pointer, and on release what it caught becomes the selection. The keys are the 3D view's box-select, to the letter — a small badge beside the pointer shows them while you drag:

Hold	The box...
nothing	catches whatever it touches () and replaces the selection
	catches only what lies fully inside it ()
	adds what it caught to the selection (green +)
	removes what it caught from the selection (red -); Alt wins over Shift

 combines with  or . "Touches" means the item's true outline, not its bounding box: a box drawn in the empty corner beside a diagonal arrow or a rotated rectangle does not catch it, and a masked image counts by the window you can see.  while dragging drops the box and leaves the selection as it was. A press and release without a drag is still a click on empty space (it deselects; with  or  held it leaves the selection alone). The box selects overlay items only — header items, which live on every step, are added on purpose with -click.

Moving several items together.  - or -click adds items to the selection — overlay items and header items can be selected together — and dragging any one of them moves the whole selection as **one rigid group**: the items keep their places relative to each other however fast the drag starts, and the move is **one** undo step. What stays put inside a selection: pinned items and constant text boxes (a status line says how many) and 3D-anchored arrows, whose place belongs to the model. An interface panel in the selection brings its bonded shapes along. The group follows the verdict of **the item you hold**: drop it over an interface panel and every carried shape bonds to that panel, drop it over none and they are set free — except shapes whose own panel travelled with them, which simply stay bonded. A carried linked shape updates its link like a dragged one. Holding  or  while pressing on a selected item takes it *out* of the selection first, so it then drags alone. While a pin is armed for **Set as new position**, the drag moves the pin only.

 **The magnet.** While you drag an item, its left edge, centre and right edge — and its top, middle and bottom — **stick** to the same lines of the other items, of the header items, and of the picture itself (its four edges and its centre). A thin magenta guide shows what it is standing on; two guides at once mean it sits on a vertical and a horizontal line. **The part you hold is the part that sticks**: hold an item by its left edge and the left edge snaps, hold it by the middle and its centre does; only when that line finds nothing nearby do the others get their turn. A selection of several items moves — and snaps — as **one box**, by its outer edges and its centre, so the group keeps its shape. Hold  while dragging to let go of the magnet for that moment (fine placement right next to a line). The **helper row** under the overlay bar holds  **X-ray**,  **Magnet** ( switches it on and off) and a  that opens the helpers' settings: what the magnet sticks to (other items / the picture) and how far it reaches (in screen pixels, so the pull feels the same at any zoom). These are *your* settings on this machine, not the project's; nothing of the magnet is rendered or saved. 3D-anchored arrows are not snapped to (they follow the camera), and a masked picture counts by the window you can see.

🔪 **Lines and arrows: points, heads and bends.** A line and an arrow are the same thing — a line is an arrow that wears no head. **Click** one and it is a unit: move it, turn it, scale it. **Double-click** it and you are editing its **points**: a dot appears on the head, on the tail and on every point between.

- **Drag a dot** to move that point. The magnet works for points too: a point sticks to the other items, to the picture, and to the *other points of its own line* (so a bend lines up with an end) — never to another line's points. **Shift** keeps the point on one axis, **Alt** lets go of the magnet.
- **Double-click the line** where there is no dot: a new point, right there.
- **Double-click a middle dot**: that point is deleted.
- **Double-click an end dot**: the arrowhead at *that* end goes on or off (a filled end dot wears a head). Both ends are independent — two heads, one, or none; give a line's end a head and it has become an arrow.
- Drag the line itself and the whole thing moves, dots and all. **Esc**, a click on empty space or on another item puts the dots away.

Right-click a line ▶ **~ Bend** chooses how it goes through its points — the cables' three ways: **Corners**, **Smooth (Bézier)** (a curve through every point) or **Fillet (rounded corners)**, with a small / medium / large fillet radius (a corner that has no room takes a smaller radius by itself). New lines are born with Fillet, so the first point you add bends the cable way. Every point move, added or deleted point, head and bend is one undo step; linked lines carry their points, heads and bend to every copy. 3D-anchored arrows keep their own rules (their ends belong to the model).

⬆ **Moving straight.** While you are dragging — the mouse button still down — hold **Shift**: the move stays on **one axis**, straight along X or straight along Y, measured from where the item (or the group) stood when you pressed. Whichever way you have gone further decides, so you can swing from one axis to the other; let go of **Shift** and the item is free again. It answers at once, even with the mouse standing still, and it works with the magnet on or off (with it on, the magnet then pulls along the free axis only).

Resizing a rotated group. The frame around several items always stands upright, even when the items in it are turned. Stretching that frame on one side only therefore squashes turned items *along the frame*, not along themselves — a turned circle leans into an ellipse. That is by design; hold **Shift** (or use a corner on images) to resize in proportion and keep the shapes true. One resize or one rotation of a group is **one** undo step, and undo brings back everything — size, angle and the lean.

Links, constants and pins

👁 Make linked shape... / Link to

Several items across steps share one definition: edit one and every linked copy follows. **✂ Break link** (make unique) detaches a copy; **↶ Reset this one to the link** re-syncs it; **🏷 Rename link...**

📌 Make constant text box... / Attach to constant

A text box that appears in the same place on many steps with the same style — chapter titles, part callouts. The anchor (⌞ top-left / ⌞ top-right) decides which corner stays put when the text length changes. **↶ Snap back to constant position**, **➕ Set as new position (all steps)**, **✂ Detach from constant**

(**this box only**). Constant *titles* (their texts and translations) are managed in **Edit ▶ Title Manager...** (12.5).

Make pinned position... / Pin to position

The same idea for images, shapes and video: a named position in the frame. A selected pinned item (and a constant text box) wears a red pin at its anchor corner — on the crop window when the image is masked. It cannot be dragged or nudged: the definition owns its position. Trying opens a small bubble on the item.  **Reposition (all steps)** in that bubble (or right-click ▶  **Set as new position (all steps)...**) lets you drag or nudge it freely; the bubble follows and offers  **Set new position** (the definition is rewritten, every step follows, undoable) or  **Reset to original**. Leaving the step before choosing cancels the move and says so in the status bar. To move only one copy, right-click ▶  **Unpin (this item only)**.  **Snap back to pinned position** returns a stray copy to the definition.

Crop masks

A mask crops an image or video to a rectangle fixed on the canvas (a window through which the picture is seen). Right-click an image ▶  **Add mask...** ▶  **New rectangle mask...** (private to this image) or  **Use a shared mask...** from the project library.  **Edit mask** (moves every image using a shared one),  **Remove mask (show the whole image)**,  **Use another mask...**,  **Set as global mask...** publishes a private mask to the library.  **Manage masks...** opens the **Masks & pinned positions** panel, which lists every mask and pin with the steps that use them and jumps to each.

Text and shape styles (Style tab)

+ **New style** (text) and + **New shape style** capture font, size, colour, fill, outline and thickness under a name. Apply from the toolbars' style dropdown. **Save Setup / Load Setup** write and read the whole style set as a file, so a house style travels between projects. Changing a style re-styles every item using it and stars the steps where those items are.

Interface panels

An *interface* is a reusable image panel (a control panel, a display) whose buttons are traced as bonded flat shapes and whose crops act as zoom windows. Interfaces are authored from the overlay and the  tab; their library folder and default pose are stored in the project. Ask for the interface guide if you need this feature — it has its own workflow.

11. Headers

The header is the strip of live text drawn over the picture on every step: step number and name, chapter number and name, free text, an image (logo), a progress bar, and the subtitle line. It is one shared layout, evaluated per step.

Header tab

Add items with + **Step #**, + **Step Name**, + **Chapter #**, + **Chapter Name**, + **Project Name** (the project file's name, so name the file after the procedure), + **Text**, + **Image**, + **Progress Bar**,  **Subtitle**. Each item has X / Y (px), Width / Height, Font size, Color, Direction (L / C / R alignment), Language (which language's text it shows) and, for images, **Replace image...** Reorder with ↑ ↓, hide with the eye, delete with ✕. **Save Setup / Load Setup** store the layout as a file.

The **Text** item's placeholder reads "Header text (used in Default / template modes)": it is the fallback when a step has no own text; it can be translated with the  button once language packs exist.

Brand — the company standard in one file

Tools ▶ Brand... manages a `.sbsbrand` file: the header items and logo, the header default, text styles, shape styles, constant-title positions, pinned positions and shared crop masks of a company.  **Save brand...** writes it from a project that looks right (and links the project to it); on a linked project it offers to **update** the brand — the revision number goes up — or to save a new one.

 **Load / update from a brand...** brings a project to the standard. Definitions already linked to the brand are **updated in place**: their identity inside the project does not change, so every text box bound to a style, every constant title and every pinned item simply follows — a new logo or colour scheme reaches the whole project in one step. What the brand adds is added; what the brand no longer carries is kept and reported. The project's own definitions (not part of any brand) are never touched.

A project that was **not** born from the brand first goes through the **matching wizard**, one page per kind of definition (text styles, shape styles, constant-title positions, pinned positions, crop masks, header items). The brand's definitions are on the left; the project's own are orange chips. Drag each chip onto the brand definition it really is, or pick it from the chip's **Move to...** list. Several chips on one brand definition are **merged** into it: every text box, shape or image that used them follows, and definitions that referenced them are re-pointed. **Keep as the project's own** leaves a definition outside the brand for good; **Delete** accepts only definitions nothing uses. A same-name match is pre-filled; a merely similar one is offered as a **looks like "..."** → hint you click if you agree. Positions and masks are drawn on a small frame — the brand's mark in green, the project's in orange — so they can be matched by where they are rather than by their names. After the wizard the project is linked, and later brand updates are silent.

Everything is listed before and after in a preview, definitions you edited locally since the last load are flagged (one button keeps them), and the update is a single undo entry. Positions are scaled when the brand was authored at another frame size. The old `.sbsheader` setup files load here too.

Changing a brand element inside a project. A text style, shape style, constant title, pinned position, crop mask or header item that came from the brand can still be edited in the project — the edit stays in *this* project. SBS tells you when that happens ( *Text style "Title" belongs to the brand "ACME"...*, once per element per session), because other projects will only get the change after **Tools ▸ Brand... ▸  Save brand...** updates the brand file. If you close the window with such changes still unsaved to the brand, SBS asks first: **Save the brand...** (the window stays open; save, then close again), **Close — keep them in this project only**, or **Cancel**. Only changes made in the current session are asked about, and an undo that restores the element clears the question by itself. The unsaved-project question, if any, comes after it.

When a linked brand file has a newer revision on disk, SBS says so after the project opens, and the panel shows **↑ Update to the new revision**. Steps whose look changes get a  as with any other definition edit.

Hiding a header on some steps

Right-click a header element in the viewport ▸ **hide on this step or on the selected steps**. The hide list is stored on the step, so it travels with copies and imports. The eye in the Header tab hides an item globally instead.

Headers in export

Headers are composited after the 3D render, so editing a header never invalidates the 3D cache; only the stitching is redone. The subtitle item shows the current narration sentence, timed from the synthesised audio.

12. Narration, voices and subtitles

Writing the narration

Type in the field under the viewport. The text is stored on the step. Pasting from a document is sanitised (smart quotes, invisible characters and paragraph breaks that upset the voice engine are normalised). A change to the text starts the step for re-render because its audio and duration change.

Local voices

SBS ships a neural text-to-speech engine that runs on your GPU (WebGPU) with an automatic CPU fallback. The voice is chosen per language in **Edit ▶ Languages...** ( click to choose). The **Export** tab has **Voice** and **Speed** for the active language. Clips are cached under `audio/<lang>/<voice>/`; **Clear inactive voices** deletes the subfolders of voices you no longer use (safe — they re-synthesise on next play), **Clear all cache** wipes everything.

Slow or silent voice? Open Developer Tools (`Ctrl` + `Shift` + `I`), Console, and run `await window.sbsTTSDiag()`. It reports the engine, device, warm-up state and the last error in plain words. See chapter 21.

Cloud text-to-speech (optional)

Settings ▶ Cloud TTS accepts a Google Cloud Text-to-Speech API key. When set, cloud voices appear in the voice picker. Audio still lands in the same cache folder, so exports are offline once synthesised.

Subtitles

Add the  **Subtitle** item in the Header tab. Subtitles are split into sentences and timed from the audio. The narration of other languages comes from their language packs (chapter 13).

Title Manager (constant titles)

Edit ▶ Title Manager... manages the texts that repeat across steps:

-  **Constant titles:** each constant text box type, how many instances it has, and   to jump to the previous / next step using it. **Delete every type with zero instances** tidies up; the menu item **Clean Up Unused Constant Titles** does the same.
-  **Translated titles:** side-by-side original and translation per language;  **Accept translation**,  **Update to translated version**,  marks rows changed since you last looked.

-  **Replace rules:** find / replace rules applied to titles and narration in one language.  **Bake** applies every rule and then deletes the rules — the corrected text becomes the text.

Edit ▶ Unify Constant Titles ( +  + ) merges text boxes that carry the same text into one constant, so a later edit changes them all.

13. Languages

One project, many languages. The project file keeps the **source language**. Every extra language is a sidecar file `<project>.<lang>.sbslang.json` holding that language's narration, header texts, overlay texts and constant titles, plus its own audio and render folders.

Edit ► Languages...

- **+ Add language** creates a pack (for example `he`) and translates the whole project into it with the engine chosen in **Settings ► Translation engine** (an OpenAI-compatible server URL and model name, with a glossary of terms that must not be translated; parallel requests configurable).
- Click a language to make it **active**: the viewport, narration field, header and exports switch to it. The status bar shows the active language.
-  chooses the voice for that language;  re-translates the parts that changed in the source;  removes the pack (the sidecar file is deleted after confirmation).
- **Settings ► Preferred narration languages** orders the languages the voice picker offers first.

What is per language and what is shared

Per language (in the pack)	Shared (in the project)
Narration text, subtitles, header texts, overlay text boxes, constant titles, replace rules, voice choice, audio cache, render cache	Model, tree, steps, cameras, colours, overlay geometry (positions, images, shapes), header layout, animation, hardware, cables, notes

Editing a text in a non-source language edits only that pack. Editing the source marks the corresponding rows in the other packs as stale ( in the Title Manager and in the Languages panel) so you can re-translate just those. Exports are made per language; the render cache is per language too, so a text-only change in Hebrew never re-renders the English video.

Translation and proofing sheets (Excel)

For a translator or a proofreader who works outside SBS, **Tools ► Review form...** opens a panel with  **Create sheet...**,  **Import sheet...** and the list of review notes. Both directions work while the project shows its original language.

Export writes one `.xlsx` workbook into the project's `exports/` folder (you can choose another place) with **one tab per language** you tick — each translation language, and optionally the original as a proofing sheet whose Target column starts empty. One row per line of text, in timeline order: a **Chapter** row wherever a chapter starts, then each step's **Voiceover**, **Step name**, **Title** and **Text box** lines, and the project **Header** texts at the end. Every step gets a Voiceover row even when it has none yet, so the client can write one. The **Step** column shows the number exactly as the Step Number header shows it (per-

chapter restart honoured, sub-steps as 5.1, 5.2). Hidden steps and hidden chapters are left out unless you tick **Include hidden steps**. **Source** holds the original text, **Target** the current translation, **Notes** is free for remarks back to you. Optionally a **Preview** picture of every step sits next to its voiceover — freshly rendered at 300 px (SBS walks through the steps once) or the small step thumbnails. Rows are colour-banded so a step reads as one block: chapter rows blue, steps alternating two tones, a line where each step starts. The visible **key** column is the step's permanent identity (locked; a hidden fingerprint column sits next to it). The sheet is protected: only Target and Notes accept typing; rows may still be sorted and filtered.

Import accepts `.xlsx` and OpenDocument `.ods` (what Apache OpenOffice and LibreOffice save natively; the old binary `.xls` is not read — choose "ODF Spreadsheet" or "Excel 2007+" when saving). It reads every language tab of a returned workbook, one after the other, and matches every row by the hidden **key** column — the step's permanent identity, so it does not matter that steps were added, moved or renumbered while the client worked. If the key column was deleted, rows are matched by the source fingerprint, then by exact source text; a row that matches nothing is reported and skipped. A row whose source text you edited after the export is flagged  "source changed after the export", and one button applies only the rows that are still safe. Empty Target cells never erase an existing translation. A sheet from another project asks before proceeding. Everything is listed before and after in a preview, lands as one undo entry per tab, and — for a target language — goes into its pack; switch to that language to see it. A proofing sheet of the original changes the project text itself, and the other languages then show those lines as drifted.

Review notes (tab)

Whatever the client typed into **Notes** — and any picture they floated over a row, a screenshot for example — comes back as a **review note** tied to that step. A step's several rows (voiceover, name, titles) fold into one note, each cell labelled by its row type, pictures collected. The  tab next to **Export** lists them: a tick marks a note as addressed, the filter shows open, addressed or all, and clicking a note jumps to its step and opens the note window.  **Open note window** opens that movable window on its own: full text, the pictures,   through the list, **Addressed**, **Go to step**, delete. Notes are saved in the project;  **Clear addressed** removes the ones you have dealt with.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: an exported sheet open in Excel — chapter rows, step numbers, previews, Source and Target. File: `docs/manual/img/translation-sheet.png`

14. Hardware: screws, nuts, washers

CAD assemblies often ship without fasteners, or with hundreds of identical ones. The  tab lets you define a screw once and drop instances onto faces.

Templates

+ **Create template** opens the editor: **Head type**, **Drive style**, **Ø Diameter (mm)**, **Length (mm)**, **Thickness**, **Colour**, washer names, **Spacing X (mm)** and **Gap scale** for stacked washers, **Reposition (ms)** for the pre-insertion move. Save with **Save changes**.  edits,  deletes,  restores the defaults from **Settings** ▶ **Hardware insertion defaults**. Editing a template changes every instance made from it (the tree menu says so:  **Edit template (affects all instances)...**).

Placing

Click **Assign** on a template, then click a face on the model: the screw is placed normal to that face, head outside.  or right-click cancels. In the tree, right-click a screw for  **Duplicate (same template)**,  **Add nut** /  **Delete nut**, the washer stack ( **No washers ...**  **Spring + flat washer**) and  **Adjust insertion animation....** The **HW** filter in the tree shows only hardware.

Insertion animation

When a screw is hidden in step n and visible in step n+1, SBS animates it driving in: it first appears at the reposition distance along its axis (over **Reposition (ms)**), then travels in over the **insert** channel. Nuts and washers follow the same rule in reverse order. Hide a screw in a later step and it backs out.

15. Cables and sockets

A cable is a smooth tube between two sockets with a controllable path. Sockets are small blocks you attach to parts; cables plug into them. Both are animated between steps.

Creating

 tab ▶ + **New Cable**: name, **Cable color**, **Diameter (mm)** (the tab's diameter field sets the default for new cables), socket **W / H / D (mm)** and **Socket color**. Place each end by clicking a face (the placement icon on the row), then add mid-points by clicking along the path. Duplicate a cable with **Duplicate** and choose **Keep connections** or **Clean connections**.

Cables in steps

Each step stores the cable's point positions and which socket each end is plugged into, so a cable can be unplugged in step 12, dangle in step 13 and be plugged into another socket in step 14. Editing points in a step changes only that step; the eye and highlight toggles on the row are per step too. When the part a socket is attached to moves, the socket and the cable end follow. A step's context menu lists the recorded cable actions (**cable · plug/unplug · socket · State n**).

Sizes are absolute millimetres. Projects made with the older percentage sizing are converted on load so they look the same.

16. Flat shapes and primitives

Flat shapes (■)

A flat shape is a filled outline traced from a model face: highlight a gasket surface, mark a sealing area, show a footprint. Click the trace button, then a face — adjacent triangles within **Crease angle (degrees)** of the picked triangle's normal join the face set and their outline becomes the shape. Tighter = only flat regions; wider = catches gently curved surfaces. Shapes belong to **shape groups** (double-click to rename); the eye on a group toggles every instance in it; the lock makes the group select as one.

- **Edit** re-opens the polygon in the viewport editor seeded at an existing instance.  **Edit shape...** in the tree does the same.
- **Style**: fill colour and opacity, outline colour and thickness — per shape, or from a shape style.
- Textured plane: pick an image file, then click a face (or empty space) to drop a 2D plane sized to the image's aspect ratio — decals, labels, screens.
- Shapes appear, move and fade with the **shape** channel and keep their end state exactly as the step says, even when the part under them fades out.

Primitives (⬡)

Boxes, cylinders, spheres and similar helpers built inside SBS, used as stand-ins (a missing bolt, a tool, a fluid volume) or as containers. Choose a type and size in the ⬡ tab, then click a face to place it tangent ( **Esc** cancels), or  **Align to surface...** / **Align by 3 points** later. Primitives have a base face at the origin so they sit on surfaces naturally, and they can hold children like folders: drop parts inside and they move with it. Right-click a primitive:  **Paste (independent)** versus  **Paste Instance (linked parameters)** decides whether a copy shares its dimensions with the original. Sizes and quality settings live in the project and are part of what the render cache fingerprints.

17. Notes

A note is a callout attached to a part: a label with a leader line that follows the part through steps and camera moves. Right-click a part ▶  **Add Note...**, type the text, pick a size (● **Size: Small / Medium / Large**) and a template from the  tab.  **Reposition Note...** lets you click a new anchor;  **Delete Note** removes it. Notes are per step: the part's note list is stored in the snapshot, so a note can exist in steps 5–8 only. The  tab keeps **note templates** (style, size, leader look) and the list of notes in the active step. Note text is translated with the language packs.

18. Environment and look

Env tab

- **Theme:** **Light / Dark** for the interface; **Default background** in Settings for the viewport colour of new projects, and the background picker here for this project.
- **Grid** on/off, **Outline** on/off with **Outline color** and the **Small / Medium / Large (px)** thickness presets. Outlines are a silhouette pass around parts; per-preset override in the Colors tab.
- **Lights:** **Key light (warm)**, **Fill light (cool)**, **Rim / backlight** — intensity and angle. The light stage window lets you drag them around the model.
- **Load an environment from a .sbsenv file** — or grab it straight from another .sbsproj: copy a whole look between projects.

Production render settings

The render settings panel (from the Env tab) controls what the export looks like: studio presets (**Built-in studio (default)**, **Studio — bright**, **Studio — soft boxes**, **Photo studio — warm / earthy**), **Environment strength** and **blur**, **Exposure**, **Contrast** and **Saturation (grade)**, ambient occlusion (**Intensity**, **Radius (px)**, **Steps**, **Thickness**, **Distance falloff**, **Max distance**) and the rim light spread. Changing any of these changes every frame, so it starts all steps.

Preview versus export. The viewport shows the same renderer as the export, at the viewport's size. Exports add supersampling per frame, so edges are cleaner than on screen.

19. Export, render cache and stars

Export tab

Field	Meaning
From / To	Step range to export. Leave empty for the whole project.
Width / Height (px), Format preset	Output size. Cache clips are rendered at this size; changing it re-renders everything.
Frame rate (fps)	Typically 30. Part of the cache key.
Step hold (ms)	Extra dwell appended to each step after its narration. A pause <i>within</i> a transition is authored in the animation preset (pause channel).
Voice · Speed	Narration voice and rate for the active language.
Output format · Quality	MP4 (H.264) or WebM; the quality slider is the encoder bitrate cap (variable bitrate — static content uses less).
File name · Choose folder...	Defaults to <code>exports/</code> inside the project.
Alpha masks	Also write a per-frame alpha (matte) alongside, for compositing in another tool. Off by default.
.sbsproc	Export a single self-contained file (MP4 + step manifest) for the SBS viewer.
Start Export · Cancel Export	Run / abort. Progress and the current step are shown in the status bar.

Settings ▶ **Export defaults** sets the defaults for new projects (output format, frame rate, step hold).

How the render cache works

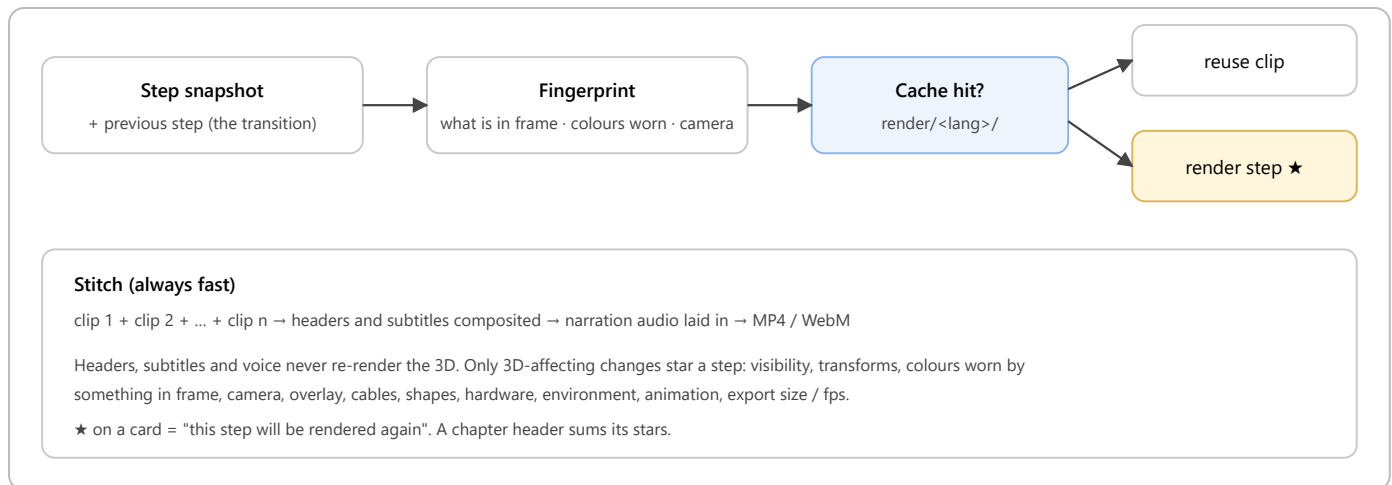


Figure 6 — Each step is a clip in the cache. Export = render the starred clips, then stitch everything.

The first export renders every step. From then on, **Start Export** opens a small prompt:

- ★ **Re-render starred** — render the starred steps, reuse the rest. The normal choice.
- **Trust the stars** — if the cache keys changed for a technical reason (an app update changed the fingerprint format) but nothing in your project did, adopt the existing clips instead of rendering. Use it once after such an update; if a clip then looks stale, star the step and re-render.
- + **current** — also re-render the active step even if it is not starred.
- ► **Render full project (incremental)** and 🎯 **Re-render selection** with a range like 0-10, C11, 124, 89-102 (C11 = chapter 11).
- 🧹 **Purge stale cache** — delete clips no step refers to any more (old sizes, deleted steps). Never deletes clips that are in use.

Stars: what changes what

A step gets a ★ when something that affects *its rendered frames* changed since it was last rendered. SBS compares against the rendered version, so undoing the change removes the star again. The rules:

You change...	Stars...
Visibility, a transform, a colour assignment, the camera, the overlay, narration text, hold, easing, private animation of step n	Step n, and step n+1 (its transition starts from n).
A colour preset's values	Every step where a part wearing that preset is in frame (see the frame scan below). Presets nobody wears in frame star nothing.
A text style, shape style, linked shape, constant title, pinned position, crop mask	Every step whose overlay uses it.
A camera template	Every step bound to it.
A cable's look (colour, diameter, style), a primitive's size or quality, a flat shape	Every step where it is visible.
Step order, chapter membership	The steps whose predecessor changed.
Animation preset (project default), AL1 / AL2 defaults, environment, export size, fps	All steps.
Header layout, subtitles, voice, narration speed, language texts	Nothing — composited at stitch time.

Deliberately not starred. Editing a hardware template, a note template, or the geometry of a loaded model (relink, source transform) does not star steps automatically. After such edits, right-click ►  **Re-render cache segment** on the affected steps, or clear the cache.

Cleared the star by mistake? ☆ **Clear star** only tells SBS to keep the current clip. If the clip looks wrong, right-click ►  **Re-render cache segment**.

Scan steps for what is in frame

Edit ► Scan steps for what is in frame... walks through every step with its own camera, at the export aspect ratio, and records which parts actually appear in the picture (ghosted and transparent parts count as visible). The record is saved next to the render cache and has two uses:

- Stars become precise: changing a colour preset stars only steps where a part wearing it is really on screen, not merely un-hidden somewhere behind the camera.
- Cache keys become narrower: a change to something off-screen no longer invalidates the step's clip.

The scan takes a moment per step (it is a GPU pass) and reports progress in the status bar. Records are refreshed automatically for steps you leave after editing them and at the end of every export; run the menu command after large reorganisations or when you want the stars to settle before a big export.

Exporting other languages

Make the language active in **Edit ▶ Languages...**, then export. Each language has its own render cache folder, so switching languages does not throw away work; only steps whose overlay text differs are rendered per language.

Document — the animation as a printed manual (PDF)

Tools ▶ Document... turns the approved animation into a paged A4 document: every page covers one or more steps, with their text on top and a picture of their final state below, inside a fixed page template with a header and a footer. The document lives inside the project file — there is nothing separate to keep in sync by hand — and it keeps its own copy of everything: **nothing you do in the document changes the animation**, its steps, its step groups, its numbering or its voiceover.

The document workspace

The document opens as a workspace that takes over the whole window; the animation's shortcuts are silent while it is open. Ctrl+S still saves. Ctrl+Z / Ctrl+Y work on **document edits only**: when the next thing to undo is an edit you made to the animation before opening the workspace, undo stops and says so, rather than changing the animation out of sight. ◀ **Edit animation** (top-right) closes it — nothing is lost; on the animation side the same corner holds **Edit document ▶** (above the steps panel), so the two buttons work like a toggle between the animation and its document. **Drag the separators** between the three panes to give the page more or less room (the widths are remembered on this machine); double-click a separator to fold that pane away and again to bring it back. On a small screen the page list gives way first and the settings fold away by themselves, so the page always stays usable.

Where	What
Right	The document page by page , under its chapters. Every box shows a thumbnail of the page itself — header, text, pictures — with its printed page number. A joined page is one box with one thumbnail, labelled <i>3 steps merged (steps 2–4)</i> , and lists its steps under it as compact rows; each step row ends with an eye that leaves the step out of the document. The contents and custom pages carry a  grip: they are the only things that can be moved. Right-click for the page commands. The list scrolls; ↑ / ↓ walk the steps, ← / → (or PageUp / PageDown) walk everything that prints — the same arrows that step through the animation.
Centre	The selected step's page — A4 in true proportion, header and footer included. It is the very page the PDF is printed from, so what you see is what prints. Fit / 100% set the zoom.
Left	The document's fields (title, company, document number, revision, numbering) and the selected page: its template, its pictures, its  marks and the state of each of its text lines.

Pages

- **Build the pages — one per step** is the starting point. A step with sub-steps counts as one unit (it plays as one clip); its sub-steps become extra lines on the same page. Hidden steps and hidden

chapters are left out.

- **Selecting.** A selection is always **one unbroken range**: click a step, then Shift-click or Ctrl-click the other end — everything in between is selected. There is no way to leave a hole in a range, so a join can never leave a step outside.
- **Joining.** With a range selected press ↩ **Join steps 25–28 into one page** above the list, or right-click the selection and pick the same command. The joined steps keep **the animation's own numbers** — 25, 26, 27, 28, not 25.1, 25.2 — because the manual and the animation must agree on what "step 27" is.
- **Splitting.** Select one step inside a joined page: ⏮ **Start a new page at step 27** (button or right-click). **Un-merge — one page per step** undoes a join completely.
- **Leaving a step out.** Click the 👁 at the end of a step's row (or right-click ▶ **Leave ... out of the document**; with a range selected it applies to the whole range). The row stays in the list, dimmed, with a shut eye 🙈; click it again to put the step back. This is a document-only decision: the step is **not** hidden in the animation, the other steps keep their numbers, and a page whose steps are all left out is simply not printed.
- **The layout follows the number of steps.** A page is on *Automatic* until you choose otherwise: join two steps and the page gets two pictures, three steps → three, four or more → four (sort the rest out by hand), always in step order; un-merge and the pages go back to one picture. Pick a template yourself and the page keeps it — choose **Automatic** again in the template list to hand it back.
- Each page has a **template**: *One picture — wide (16:9)*, *One picture — tall (3:2)*, *Two pictures side by side*, *Three pictures — one large, two below*, *Four pictures — two by two*, or *Large picture, short text*. Templates are strict on purpose: a picture can only live in a template frame, so every page of the manual lines up.
- **Pictures are cropped into their frames.** A frame does not need the video's proportions: the picture fills it and what does not fit is cropped. **Click a picture** to select it — then **drag** to move it behind its frame and use the **mouse wheel** to scale it about the pointer (one undo step per drag). The bar above the picture has **Fill** (back to the default crop), **Whole** (the entire picture inside the frame), – / +. If you make the picture smaller than its frame the white page shows around it — that is allowed. Esc deselects.
- **Step numbers on the pictures.** On a page that prints several steps every picture carries the number of the step it shows (the same number as its line), so the reader can match text and picture. Switch it off with **Step number on each picture** in the left pane.
- **Before and after.** A step picture normally shows the step's *end*. When the end hides what happened — a part that is already inserted — add the **before** frame: **Picture ▼ ▶ ⏮ before step 27**. It shows the state the step *starts from* (the previous step's end) seen through *this* step's camera, so the pair is framed identically; it carries no overlay or notes (those describe a step's end) and is labelled *27 before*. Put the before-frame and the step's own picture side by side in a two-picture page.
- **Which picture.** By default every frame is *Automatic*: the page's steps are shared out so that the **last frame always shows the page's final state** — four steps on a two-picture page show steps 3 and 4, on a four-picture page steps 1 to 4. **Picture ▼** on the picture's bar — or the **Picture 1 / 2 / ...** button of that frame in the left pane — opens a **picker with thumbnails**: every step of the page with its picture, number and name, under each one its ⏮ *before* frame (tagged *before*; until it has been rendered the thumbnail is the previous step's, which shows the same state), and the current choice is marked. It lets you choose a specific step of the page, leave the frame empty, or take an **external**

image — a photo or a drawing that is not part of the animation. An external image is scaled down to at most 2400 px and stored inside the project. A step picture is the step's export frame exactly as the video shows it — the step's recorded camera (even while Work Camera is on), 3D, overlay, notes and hardware tags — but **without the animation's header items and subtitles** (the page has its own header and its own text), and without selection highlights or other authoring marks. Step pictures are rendered when a page is shown or exported (the app walks the needed steps behind the workspace and returns to where you were); they are not stored in the project, so the file does not grow. A picture refreshes by itself when its step, its overlay, a colour or a style changes; after anything else (a reloaded model, render settings) use  **Render the pictures again** in the left pane.

- **Numbers** in front of each line: the same step numbers as the animation (default), or 1, 2, 3 on every page, or none.

Chapters and the table of contents

Every page shows the name of its chapter at the top of its text, so a reader who opens the manual anywhere knows where he is. On the chapter's first page the name is a real heading — that is what the PDF's bookmark points at — and on the following pages it looks the same without creating more bookmarks: one bookmark per chapter, however many pages it has.

The document opens with a **table of contents** of its own (the video's time-code contents has no meaning on paper): one line per chapter that actually prints — chapter number, name, a dotted leader and **the page the chapter starts on**. It is built by itself and stays right by itself: join or split pages, leave steps out, sync with a changed animation — the page numbers follow. A chapter whose steps are all left out of the document has no line. In the PDF every line is a link to its page. In a Hebrew document it reads תוכן עניינים and runs right-to-left.

- The contents is a real page and is counted: at the front it is page 1 and the first step page is page 2 (*Page 2 / 41*). With more than 26 chapters it takes a second page and everything after it shifts by two.
- **Put it where it belongs.** A contents is rarely the very first page. Drag its  grip in the list — a blue line shows where it will land — or right-click it: **Move up / down**, **Move to the start / the end**. It is anchored *before a step*, so it keeps its place when you join, split or re-sync pages. Everything is renumbered, and its own lines follow.
- In the workspace it is a box of the list,  **Contents**: click it to see it in the centre; the arrows reach it like any other page. The list shows the **printed** page numbers.
- Switch it off with **Table of contents (chapters → pages)** in the left pane; the numbering then starts at 1. A project without chapters has no contents page.

For now the contents lists chapters only. Listing the steps of each chapter as well is planned, once it is clear how much detail a long manual can take on one page.

Custom pages

Not every page of a manual is made of steps: a cover, a safety notice, a parts list, a revision table. **Custom page** (above the list) adds an empty page right after the page you are on. It carries the document's header, footer and watermark and is numbered like any other page; between the header and the footer it is yours.

- The bar over the page adds items: **+** **Text** and **+** **Picture** ▾ — **From a file...** (stored inside the project, at most 2400 px; its frame starts in the picture's own proportion) or **From the animation...**: a list of **every step**, chapter by chapter, each with its thumbnail. A step picture is not stored: it refers to the step and is rendered like the page pictures, so it follows the step when the animation changes. With it selected the bar offers **↗ Step N** ▾ (another step) and **↳ Before** (the state the step starts from, seen from its camera, instead of its final state).
- **Click** an item to select it, **drag** it to move it, drag a **handle** to resize it; arrows nudge by 1 mm (Shift = 5 mm); **Delete** removes it; **↑ Front** brings it above the others. Items snap to a half-millimetre grid and stay inside the content area.
- **Double-click a text** to type; click away to finish, Esc to abandon. Text wears the document's body font — the same one as the step texts, so the manual stays of a piece — and you choose the size (any number of points), bold, italic, alignment and colour. Hebrew and mixed lines are handled per box.
- A picture is cropped by its frame like the step pictures: **Fill**, **Whole**, **–** / **+**.
- Every gesture — a drag, a resize, a typed text, a delete — is one undo step.
- The page has a name for the list (left pane), a **☰** grip to move it anywhere in the document, and a right-click menu: move, rename, delete. It is anchored before a step, exactly like the contents.

Tables, and a text box / table of this kind for the animation's overlay, are planned.

Your own page templates

When none of the prefab layouts fits, draw one. In the left pane, under the page's template, click  **New template....** The centre turns into a layout editor that starts from the current page's layout: the header and the footer are fixed grey bands, the **text box is blue**, the **picture frames are green** and numbered in the order the pictures fill them (the side panes are dimmed until you save or cancel).

- **Drag a box** to move it; drag a corner or an edge handle to resize it.
- **Drag on empty paper** to draw a new picture frame — or press **+** **Picture frame**. Up to eight frames; a template with no frame at all (a text page) is allowed.
- The bar shows the selected box's **X, Y, W, H in millimetres** — type exact values — and, for a picture frame, ratio buttons (**16:9 3:2 4:3 1:1 3:4**: the width stays, the height follows), **◀ # / # ▶** to change which picture the frame receives, and **Delete frame**.
- Keyboard: arrows nudge by 1 mm (Shift = 5 mm), Tab goes to the next box, Delete removes the selected frame.
- Everything snaps to a half-millimetre grid and stays between the header and the footer, inside the side margins. **Boxes may not overlap**: overlapping boxes turn red, the bar names them, and **Save template** waits until they are apart. A frame does not need the video's proportions — pictures are cropped into it.

Save template asks for a name, stores the template **inside the project**, puts the current page on it and adds it to the template list of every page. A page that uses one of your templates offers  **Edit** (changes every page that uses it) and **Delete it** (those pages go back to the automatic layout). In a right-to-left document your frames are mirrored like the prefab ones, so draw them in left-to-right order.

Text

Every line starts as the step's voiceover and *follows* it: change the narration and the document changes with it. To rewrite a line for print, **click it on the page and type**; clicking away commits it, Esc abandons the change. From that moment the line is the document's own copy (*rewritten for the document*, blue ring on its number) — the voiceover is not touched, and the other way round. If the voiceover changes later, the line gets an amber ring and the left pane shows **⚠ the voiceover changed after you rewrote this** with the new voiceover underneath, so you can decide: **keep mine** or **use the voiceover again**. Hebrew and other right-to-left lines are laid out right-to-left automatically, number badge included.

When a page has more text than its template has room for, the text area turns red with **✂ the text does not fit** — live, while you type. Shorten the text, split the page, or choose a template with a larger text area. Export asks before it writes a PDF with a cut page.

When the animation changes afterwards

Pages refer to steps by their permanent identity, not by position, so you can keep working on the animation. When the step list no longer matches the pages, the **☺ Sync** button turns amber and steps the document does not know yet appear dimmed in the list. Syncing never changes anything silently — every adjustment leaves an **!** on the page it touched, with a sentence saying what happened:

What changed in the animation	What sync does
A new step between two steps of the same page	It joins that page (⛶).
A new step anywhere else	It gets a page of its own, in order (NEW).
A step was deleted or hidden	It leaves its page (⇐). A page left with no steps stays, marked ∅, until you delete it.
A step was moved to another place	It leaves its page (↗) and is placed like a new step where it landed. A whole page that moved together simply moves, untouched.
A picture was pinned to a step that left the page	The picture is kept and flagged (📌) — you decide.

Read the marks, fix what needs fixing, then ✓ **Seen** — **clear these marks** (left pane) or ✓ **Clear all** **!**.

Hebrew and other right-to-left documents

Reading direction is always in the left pane, in the **Document** block under **Step numbers** — not only when the project is in Hebrew. It is *Automatic* by default: SBS counts the letters of everything the document prints, and when most of them are Hebrew or Arabic the whole document turns right-to-left — the header and footer mirror, lines and their number badges start from the right, picture numbers move to the top-right corner, and **the first picture of a page is the right-hand one** (the frames are mirrored across the page, so the reading order of the pictures matches the text). The built-in phrases print in the document's language (מהדורה 12, מתוך 1 עמוד B, לפני) unless you re-worded them. A mixed line — Hebrew with a part

number or a torque in Latin letters — is handled per line. Force **Left-to-right** or **Right-to-left** if the guess is wrong for your document.

Watermark

The left pane has a  **Watermark** section (click its title to open it). Tick **Print a watermark on every page** and choose the kind:

- **Text** — DRAFT, CONFIDENTIAL, a customer name; two lines are fine, Hebrew too. Size is in points and takes **any number**; pick a colour. The default is a mid grey because it reads both on the white page and over a dark picture — black disappears on dark pictures, white on paper. In the PDF the mark is real, crisp text at any zoom.
- **Image** — a logo or a stamp. **Choose image...** opens a conversion window, because logos arrive in three shapes and a watermark needs real transparency: *artwork on a white background* (the white is made transparent), *artwork on a black background* (the black is made transparent), or *already transparent* (kept as it is). SBS guesses which one it is and shows the result over a white page and over a dark picture so you can judge it; **Make it one flat colour** turns a coloured logo into, say, a grey mark. The conversion is exact — soft edges stay soft, colours stay true — and is done once: the converted image is what the project stores and what every output uses. **Width** is a percentage of the page width.

Opacity and **Angle** are sliders that redraw the mark on the page while you drag (one undo step when you let go). **Where:** *Over everything* — *the pictures too* (the usual choice: the mark also protects the pictures) or *Behind the text and the pictures*. The watermark never gets in the way of editing: clicks go through it to the text and the pictures. It belongs to the document only — the animation and the video are not affected — and an image watermark is embedded once in the PDF however many pages there are.

Designing the header and the footer

Out of the box the header shows the logo (the project's first header image), the company, the title and *document no. · revision*; the footer shows the project name, the chapter and *Page n / N*. To design them yourself, **click the header or the footer on any page** — or  **Edit...** under *Header and footer* in the left pane. The page behind dims (it is only there to show the result; ← / → still walk the pages) and the band you are editing gets a dashed frame. The first time, the standard header and footer simply become movable items that look exactly the same — nothing changes until you change it. What you design is printed on **every page**: step pages, the contents, custom pages.

- **Header / Footer** on the bar choose the band — or click an item of the other one.
- Items work as on a custom page: click, drag, handles, arrows (1 mm, Shift 5 mm), **Delete**, **↑ Front**; text has size, bold, italic, alignment and colour and is centred vertically in its box. Items cannot leave their band (header 12–198 × 10–28 mm, footer 12–198 × 275–287 mm).
- **+ Field ▾** adds a value that fills itself in on every page: title, company, document no., revision, project name, chapter name, chapter number, page number, number of pages, date of the export. With a text selected the field is added *to that text*, so one box can read `Page {page} / {pages}`. On the page you see the values; **double-click** the text and you see what was written — the `{names}` — to edit it.

- **+ Picture ▾** adds **the project's logo** (it follows the project: change the logo there and it changes here; always shown whole) or a picture from a file (a certification mark, a second logo).
- **_ Line** shows or hides the rule of the band.
- On a right-to-left document the whole design is mirrored by itself — you design it once.
- **✓ Done** or Esc goes back to the page. **↶ Standard** (or *back to the standard one* in the left pane) throws the design away; undo brings it back.

The PDF

Title, company, document number and revision (left pane) feed the header's fields. **↓ Export PDF...** writes the whole document (into the project's `exports` folder by default): real, selectable, searchable text — not a picture of text — with a bookmark per chapter. Every edit in the workspace is undoable with Ctrl+Z.

20. Settings, maintenance and housekeeping

Settings (Ctrl+,)

Section	Contents
Keybindings	Rebind the single-key actions (work camera, overlay edit, global mode, gizmo local/world, capture camera, overlay x-ray): click, then press the new key. Default restores the factory key.
Auto-backup / Autosave	Interval, how many backups to keep, Back up now .
Hardware insertion defaults	Nuts, spacing, gap scale, reposition pre-step (ms) used when a template does not override them.
Cloud TTS	Google Cloud Text-to-Speech API key; Test .
Preferred narration languages	Order of languages offered first in pickers.
Camera	Field of view and orbit defaults for new projects.
Default background	Solid colour for new projects' viewport.
CAD import (STEP / IGES)	Tessellation quality and default tree structure for imports.
Export defaults	Default output format, frame rate, step hold.
Translation engine	Server URL (OpenAI-compatible), model name, glossary, parallel requests.

Maintenance commands (Edit menu)

Recover stuck inputs

If a text field stops accepting keystrokes (rare, after certain popups), this releases it without a reload.

Rebuild Cascade

Recomputes the folder-to-part inheritance for every step from the stored data. Use after a big tree reorganisation if some step shows a stale hierarchy.

Clean up folders...

Finds wrapper folders with a single child across the loaded models, previews the flattening, and offers **Save first**, then **remove** or **Remove without saving**.

Model source transform...

Per-model scale (Metric / Imperial), rotation and offset applied at the source, with per-axis reset and **Reset all to identity**. Steps are not re-baked automatically — re-render afterwards.

Scan steps for what is in frame...

See 19.5.

Title Manager... / Unify Constant Titles / Clean Up Unused Constant Titles / Languages...

See 12.5 and chapter 13.

Reload versus restart

Ctrl + **R** reloads the app window and re-opens the current project from disk: the live scene is rebuilt from the saved data, which cures most "the picture looks wrong but the data is right" situations. Save first — the reload is not guarded. **Ctrl** + **Shift** + **R** also clears the window cache. A full restart of the application is only needed after installing a new version.

Memory

The app window has a hard memory ceiling of roughly 3.5 GB. Very large projects (hundreds of overlay images, long image sequences, many voices) approach it. Symptoms: slow undo, a white window, an export that dies mid-way. Remedies: keep overlay images small, store video as files (they are never inlined), **Clear inactive voices**, split the project into books and stitch the exports, and **Clear history** in the ↶ tab on long sessions.

21. Troubleshooting

Symptom → what to do

Symptom	Likely cause	Do this
A part is missing or sits in the wrong place in one step, but only after jumping around	The live scene drifted from the data (an old bug, fixed in V0.3.3)	Save, then <code>Ctrl</code> + <code>R</code> . If it comes back, run <code>window.sbsDiag.visibilityAudit()</code> in the console and send the output.
Project opens with an asset warning	Model file moved or renamed	Browse... to relink, or Override to open with placeholders; fix the path later via <code>Ctrl</code> + <code>B</code> .
Narration is slow, silent, or the first play hangs	Voice engine still warming up, or GPU path failed and fell back to CPU	Wait for the warm-up message; then <code>await window.sbsTTSDiag()</code> in the console. <code>window.sbsTTS.forceCPU()</code> forces the CPU path for this session.
Text field ignores typing	Focus stuck after a popup	Edit ▶ Recover stuck inputs .
Steps 55–57 (say) always show ★ even after rendering	Something on those steps changes every time the project loads	<code>window.sbsDiag.starts(stepId)</code> lists the reason per step; send it.
Export re-renders everything after an app update	Fingerprint format changed	Choose Trust the stars once.
Instant step on a chapter start plays a half transition	Old instant block format	Re-pick the easing from the dropdown on that step.
Duplicate flat shapes vanish after reload	Known, rare	<code>window.sbsDiag.shapes()</code> then <code>window.sbsFix.pruneShapes()</code> ; report the console output.
White window, or export dies mid-way	Memory ceiling	See 20.4.

Console diagnostics

Open Developer Tools with `Ctrl` + `Shift` + `I` and pick **Console**. Everything under `window.sbsDiag` only reads and reports; everything under `window.sbsFix` repairs and is undoable or asks first. Paste the

output when reporting a problem.

Command	Reports
<code>window.sbsDiag.stars() / stars('step_id')</code>	Which steps are starred and why (reference, order, definition, animation, settings, node, tool).
<code>window.sbsDiag.frameVis('step_id') / await window.sbsDiag.scanFrames()</code>	The in-frame record of a step; run the scan from the console.
<code>window.sbsDiag.visibilityAudit() → visibilityRepair()</code>	Parts whose effective visibility disagrees with the step data.
<code>window.sbsDiag.verifyHome()</code>	Objects whose live home position disagrees with the stored data.
<code>window.sbsDiag.prims() → window.sbsFix.prims()</code>	Primitive parameter health.
<code>window.sbsDiag.shapes(), cablesAudit(), cableSteps(), iface(), videos(), archived(), rmHealth()</code>	Per-feature audits: flat shapes, cables, interfaces, video overlays, archived nodes, replace-models.
<code>window.sbsDiag.fadeTrace(), animTrace(), cableTrace(), traceSelected(), traceOff()</code>	Live traces of a transition for the selected objects; turn off with traceOff.
<code>window.sbsDiag.input() → window.sbsFix.input()</code>	Stuck input diagnosis and release.
<code>await window.sbsTTSDiag()</code>	Voice engine: device, warm-up, cache, last error.
<code>window.sbsMemReport(), window.sbsFreeMemory()</code>	Heap usage by category; drop caches.
<code>window.sbsRenderPlan(), window.sbsCachePurge()</code>	What the next export would render; purge stale clips.

22. Keyboard shortcuts

Project

Ctrl + N	New project
Ctrl + O	Open
Ctrl + S / Ctrl + Shift + S	Save / Save As
Ctrl + B	Browse assets
Ctrl + ,	Settings
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + X C V A	Cut, copy, paste, select all (context-aware: tree, steps, overlay)
Ctrl + Alt + U	Unify constant titles
Ctrl + R / Ctrl + Shift + R	Reload / hard reload the window
Ctrl + Shift + I	Developer tools (console)

Viewport (rebindable in Settings ► Keybindings)

W	Work camera on / off
C	Capture the view into the step camera
F	Fit selection (or all)
L	Gizmo local / world
Space	Global mode on / off
O	Overlay edit on / off
X	Overlay x-ray
M	Overlay magnet (snapping) on / off
Shift while dragging an overlay item	Move straight along X or Y
Alt while dragging an overlay item	No magnet for that moment
Esc	Cancel tool / deselect / close popup
Delete / Backspace	Delete the selected step, overlay item or node (with confirmation where destructive)
← / →	Previous / next step
Shift / Ctrl / Alt + click	Add / toggle / invert selection

23. Glossary

AL1 / AL2

Animation-length variables (ms) that presets reference; AL1 is the camera's home, AL2 the objects'.

Archive

Remove a part from every step without deleting it.

Cascade

The inheritance of visibility and transforms from folders to their contents inside a step.

Constant (title / position)

An overlay item defined once and shown at the same place on many steps.

Fingerprint

The hash of everything that influences a step's rendered frames; the render cache key.

Frame record

The list of parts that are actually inside a step's picture, produced by the frame scan.

Global mode

Transform edits carry into every step.

Instant block

A zero-length phase pinned to the top of a step's animation that holds every channel: a cut.

Language pack

The `.sbslang.json` sidecar holding one language's texts and voice choice.

Pivot

The point a part rotates around; also the centre of its animated arc.

Preset (colour)

A named material (colour, opacity, solidness, metalness, roughness) worn by parts.

Replace-model

A node that can show one of several alternative geometries per step.

Segment / clip

The cached render of one step (its transition and hold).

Snapshot

The complete scene state stored on a step.

Solidness

X-ray strength of a preset; below 1 the part becomes see-through.

Star ★

"Changed since last rendered" marker on a step; chapters sum their stars.

Template (camera / hardware / note)

A named definition that instances follow; edit once, apply everywhere.

Work camera

A scratch camera (W) that does not touch the step's saved view.